

NHS Guidance Assistant: Design and Governance Note

A retrieval-augmented assistant over public NHS guidance. Prototype built by Sobin Sunny.

What it does

This is a working prototype that answers plain-English questions about NHS guidance documents. It finds the passages that best match your question from a trusted set of documents, then has a local language model write the answer using only those passages, and it shows you the source behind every answer. Everything runs on the machine itself, so no data ever leaves it. That was a deliberate choice, because the guidance people ask about can be personal and sensitive.

Why I built it this way, and where it stops

I chose retrieval-augmented generation because what matters here is trusted, current answers that can be traced back to a real source, not what a model happens to remember. Tying every answer to a retrieved passage is the strongest way I know to stop a model from making something up that a person might then act on. The prototype is intentionally small. The rest of this note explains how I would grow it into a production service in a responsible way.

How I would take it to production at NHSBSA

- **Cloud and MLOps.** Package it with Docker and deploy it on the approved cloud, such as Azure, with the infrastructure written as code in Terraform. Track every experiment and model version in MLflow so any change can be reproduced and checked later.
- **Pipelines and releases.** Set up automatic ingestion whenever guidance is updated, with tests and a human sign-off before anything reaches users, and keep versions of both the documents and the search index.
- **Retrieval quality.** Measure answers against a set of known questions, keep an eye out for drops in retrieval quality and questions that go unanswered, and tune the settings based on that evidence rather than by guessing.
- **Scale.** As volumes grow, move from the local FAISS index to a managed vector store such as Pinecone or Milvus, and from the local model to an approved, governed model endpoint.

Responsible AI, safety and the law

- **Grounding and honesty.** The assistant only answers from the source guidance, and it says clearly when the answer is not there rather than guessing.
- **Being able to check the answer.** Every answer shows the document and page it came from, so anyone can verify it or challenge it.
- **Fairness.** Test it against different ways of phrasing a question and different groups of users, watch for gaps that could disadvantage anyone, and keep a person in the loop for anything that really matters.
- **Data protection, freedom of information and equalities.** Privacy is built in from the start, with local processing and no data leaving the machine, clear records of sources and decisions, and governance that follows ISO/IEC 42001 AI management practice.

The stack, in one line

Python, LangChain, sentence-transformer embeddings, a FAISS vector store, a local open model run through Ollama, and both a Streamlit and a command-line interface, with a clean, testable split between the configuration, the ingestion step, the retrieval engine and the interface.

I built this prototype to show hands-on capability against the Senior AI/ML Engineer person specification, and I would be glad to walk through the code and the design in person.